

Module 8: CloudFormation Assignment

Problem Statement:

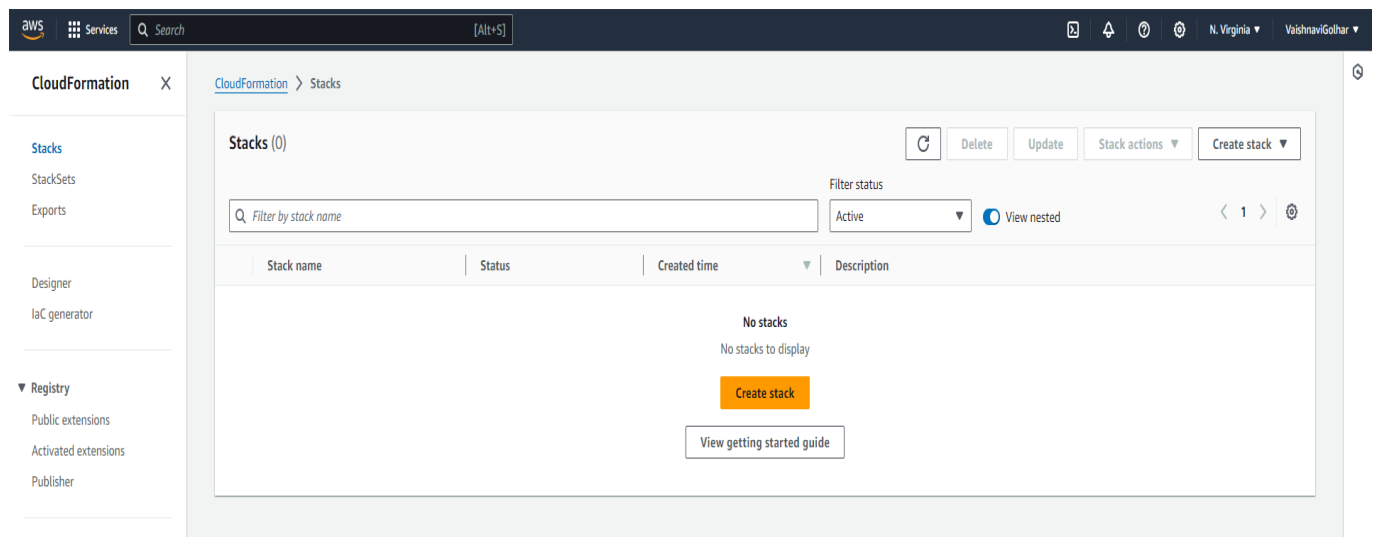
You work for XYZ Corporation. Your team is asked to deploy similar architecture multiple times for testing, development, and production purposes. Implement CloudFormation for the tasks assigned to you below.

Tasks To Be Performed:

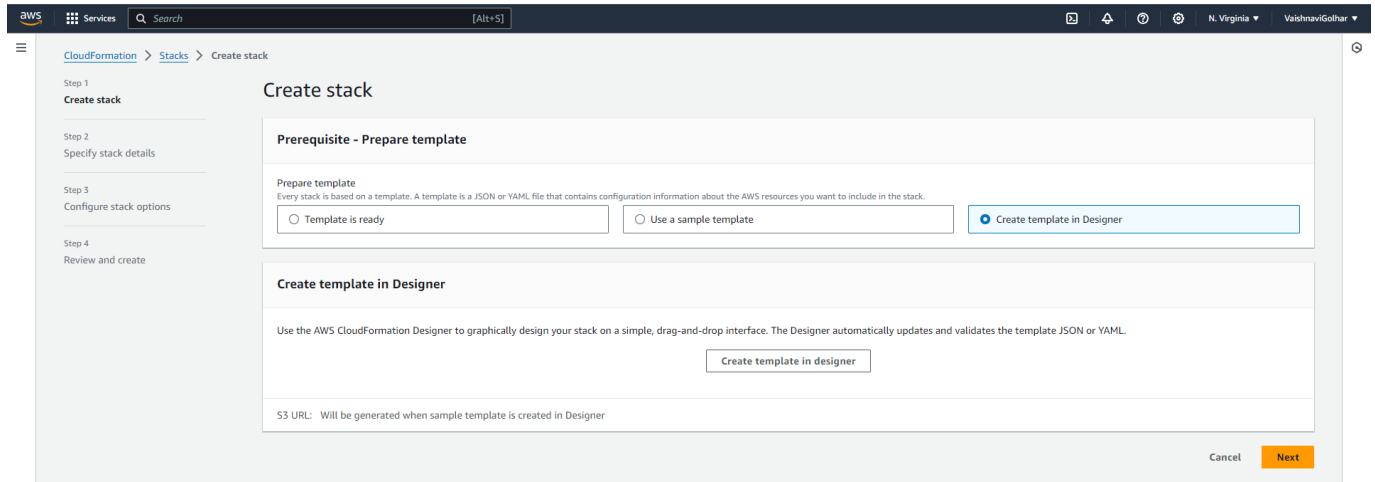
1. Use the template from CloudFormation task 1.
2. Add Notification to the CloudFormation stack using SNS so that you get a notification via mail for every step of the stack creation process

SOLUTION:

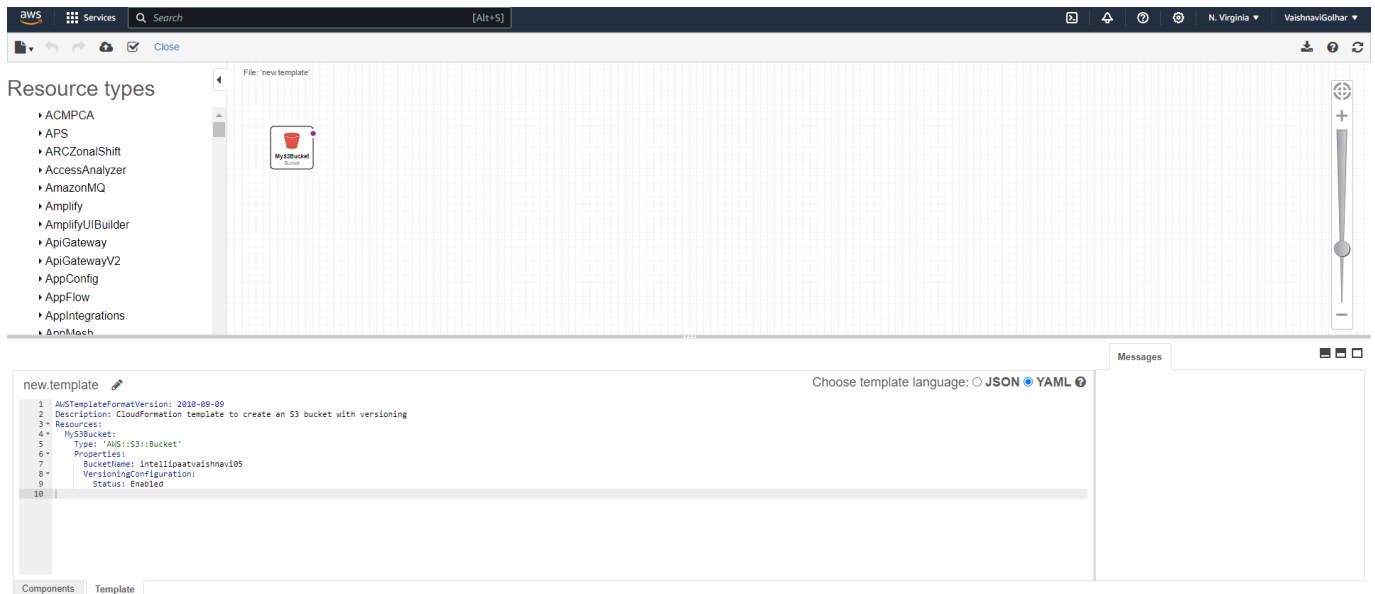
1. Click on Create stack.



2. Click on create template in designer.



3. Create a template.



4. Template will get upload in S3 bucket.

The screenshot shows the AWS CloudFormation 'Create stack' console. The left sidebar indicates the current step is 'Specify stack details'. The main content area is titled 'Create stack' and shows the 'Specify template' section. Under 'Template source', the 'Amazon S3 URL' option is selected. The 'Amazon S3 URL' field contains the URL: `https://s3-external-1.amazonaws.com/cf-templates-hsd6ivmnhfau-us-east-1/2024073nzi-new.templateai8z87ot4ls`. The 'S3 URL' field also displays this URL. At the bottom right, there are 'Cancel' and 'Next' buttons.

5. Give the name to the stack.

The screenshot shows the AWS CloudFormation 'Specify stack details' console. The left sidebar indicates the current step is 'Specify stack details'. The main content area is titled 'Specify stack details' and shows the 'Provide a stack name' section. The 'Stack name' field contains the text 'stack5'. Below the field, a note states: 'Stack name can include letters (A-Z and a-z), numbers (0-9), and dashes (-)'. The 'Parameters' section below shows 'No parameters' defined in the template. At the bottom right, there are 'Cancel', 'Previous', and 'Next' buttons.

6. Create SNS topic and attach it to the stack.

The screenshot shows the Amazon SNS console interface. At the top, there's a navigation bar with the AWS logo, 'Services', a search bar, and user information. A 'New Feature' banner is visible. The left sidebar contains navigation links: Dashboard, Topics, Subscriptions, and Mobile. The main content area is titled 'MyTopic1' and includes buttons for 'Edit', 'Delete', and 'Publish message'. Below this is a 'Details' section with a table of topic information:

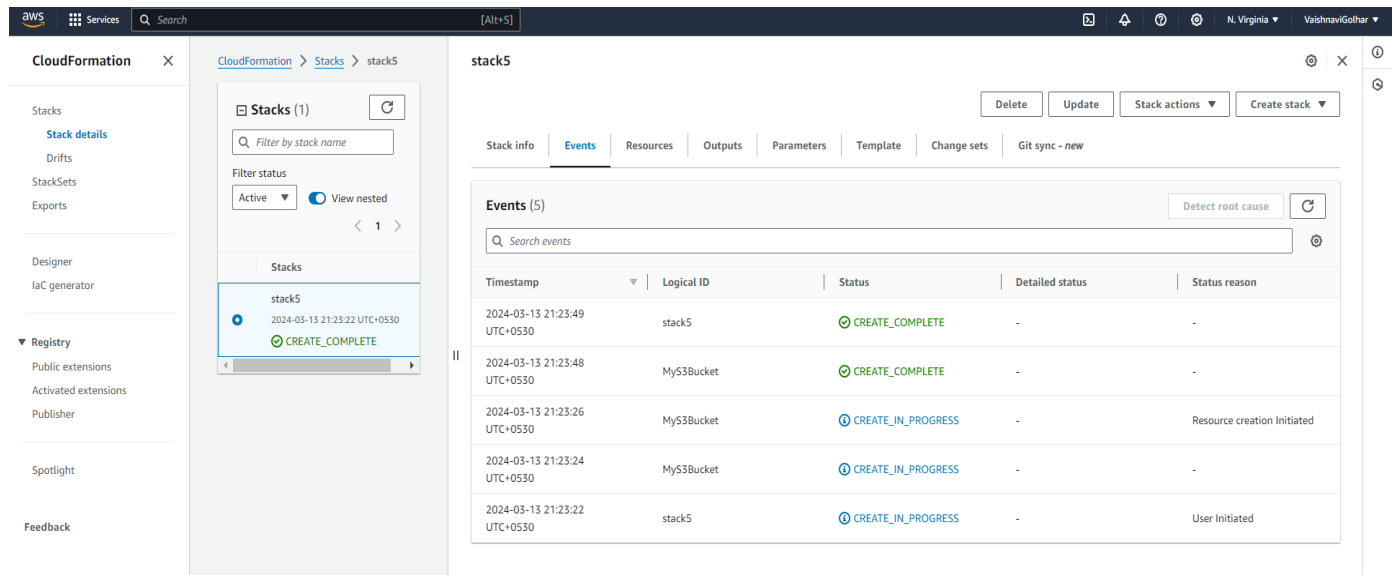
Details	
Name	MyTopic1
Display name	-
ARN	arn:aws:sns:us-east-1:161681788148:MyTopic1
Topic owner	161681788148
Type	Standard

Below the details are tabs for 'Subscriptions', 'Access policy', 'Data protection policy', 'Delivery policy (HTTP/S)', 'Delivery status logging', 'Encryption', 'Tags', and 'Integrations'. The 'Subscriptions' tab is active, showing a table with one subscription:

ID	Endpoint	Status	Protocol
66c04422-5859-4185-b492-5706ea81c487	vaishnavigolhar05@gmail.com	Confirmed	EMAIL

This screenshot shows a 'Notification options' section in a console. It features a dropdown menu labeled 'SNS topic ARN - optional' with the value 'arn:aws:sns:us-east-1:161681788148:MyTopic1' selected. Below the dropdown are two buttons: 'Add SNS topic' and 'Create new SNS topic'. At the bottom of the section is a 'Stack creation options' header. At the very bottom of the page are three buttons: 'Cancel', 'Previous', and 'Next'.

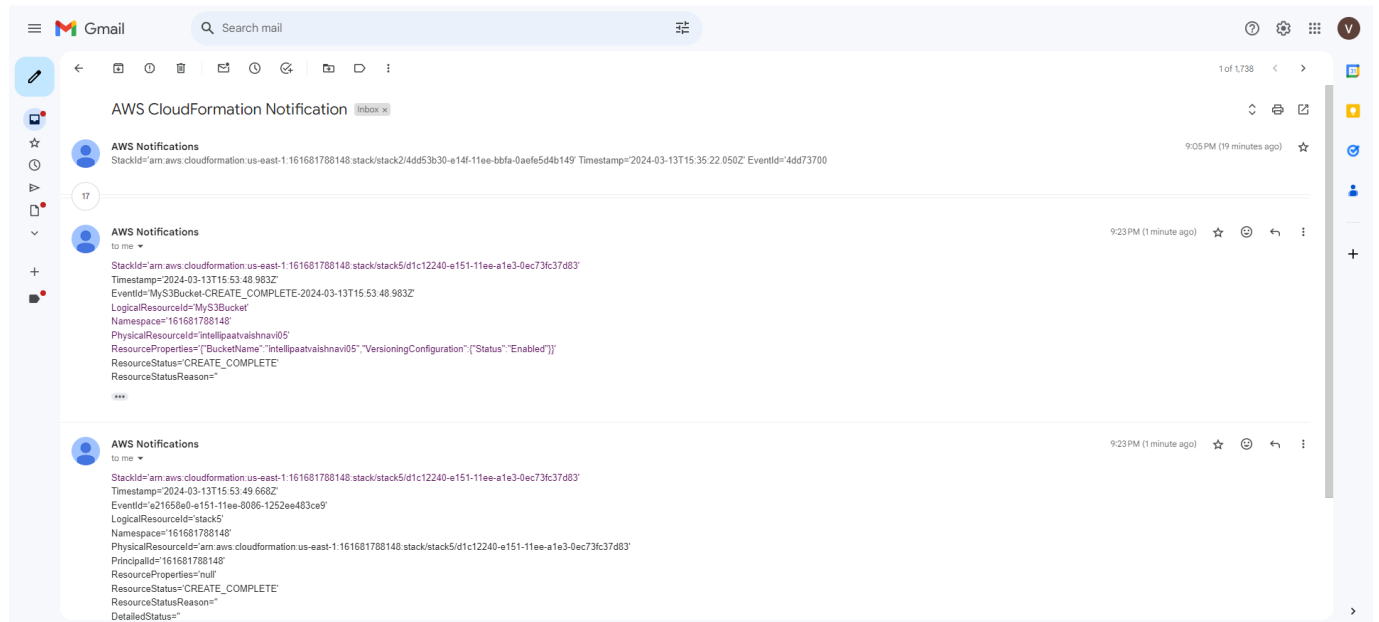
7. create a stack and wait for 2-3 min to create.



The screenshot shows the AWS CloudFormation console. On the left, the 'CloudFormation' sidebar is visible with options like 'Stacks', 'Stack details', 'Drifts', 'StackSets', 'Exports', 'Designer', 'laC generator', 'Registry', 'Public extensions', 'Activated extensions', 'Publisher', 'Spotlight', and 'Feedback'. The main area displays the 'stack5' stack. The 'Events' tab is selected, showing a list of events. The events table has columns: Timestamp, Logical ID, Status, Detailed status, and Status reason. The events are as follows:

Timestamp	Logical ID	Status	Detailed status	Status reason
2024-03-13 21:23:49 UTC+0530	stack5	CREATE_COMPLETE	-	-
2024-03-13 21:23:48 UTC+0530	MyS3Bucket	CREATE_COMPLETE	-	-
2024-03-13 21:23:26 UTC+0530	MyS3Bucket	CREATE_IN_PROGRESS	-	Resource creation Initiated
2024-03-13 21:23:24 UTC+0530	MyS3Bucket	CREATE_IN_PROGRESS	-	-
2024-03-13 21:23:22 UTC+0530	stack5	CREATE_IN_PROGRESS	-	User Initiated

8. Check email for every step of the stack creation process.



The screenshot shows a Gmail inbox with three emails from 'AWS Notifications'. The first email is titled 'AWS CloudFormation Notification' and contains the following details:

- StackId: am.aws.cloudformation.us-east-1:161681788148:stack/stack2/4dd53b30-e14f-11ee-bbfa-0aef65d4b149
- Timestamp: 2024-03-13T15:35:22.050Z
- EventId: 4dd73700

The second email is titled 'AWS Notifications' and contains the following details:

- StackId: am.aws.cloudformation.us-east-1:161681788148:stack/stack5/d1c12240-e151-11ee-a1e3-0ec73fc37d83
- Timestamp: 2024-03-13T15:53:48.983Z
- EventId: MyS3Bucket-CREATE-2024-03-13T15:53:48.983Z
- LogicalResourceId: MyS3Bucket
- Namespace: 161681788148
- PhysicalResourceId: Intellipaatvaishnavi05
- ResourceProperties: [{"BucketName": "Intellipaatvaishnavi05", "VersioningConfiguration": {"Status": "Enabled"}}]
- ResourceStatus: CREATE_COMPLETE
- ResourceStatusReason: ""

The third email is titled 'AWS Notifications' and contains the following details:

- StackId: am.aws.cloudformation.us-east-1:161681788148:stack/stack5/d1c12240-e151-11ee-a1e3-0ec73fc37d83
- Timestamp: 2024-03-13T15:53:49.668Z
- EventId: a21658e0-e151-11ee-0086-1252ee483ce9
- LogicalResourceId: stack5
- Namespace: 161681788148
- PhysicalResourceId: am.aws.cloudformation.us-east-1:161681788148:stack/stack5/d1c12240-e151-11ee-a1e3-0ec73fc37d83
- PrincipalId: 161681788148
- ResourceProperties: null
- ResourceStatus: CREATE_COMPLETE
- ResourceStatusReason: ""
- DetailedStatus: ""